

Comprehensive Edge Case Test Suite for PDF-to-Markdown Conversion

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Abstract. This document presents a **systematic collection of edge cases** designed to stress-test PDF-to-Markdown conversion pipelines. It covers structural elements, typographic challenges, mathematical notation, tabular data, code fragments, mixed-content layouts, and metadata extraction scenarios. Each section targets a specific failure mode observed in production conversions of scientific and technical documents.

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1. Structural and Layout Challenges

1.1 Hyphenation Across Line and Page Breaks

The quick brown fox jumps over the lazy dog. This paragraph contains words that are deliberately hyphenated to test whether the converter correctly reassembles them into single tokens. Common offenders include: implementation, configuration, environmental, characterization, and interoperability. The soft-hyphen character (U+00AD) is also inserted between syllables of polysyllabic words.

Dehyphenation is particularly challenging when the break occurs at a page boundary. The word on the previous page may end with a hyphen, and the continuation appears on the next page. Converters must track these across page splits.

1.2 Multi-Level Nested Lists

1. First-level item with a long description that wraps across multiple lines to test indentation handling in the output format.
 1. Second-level item
 1. Third-level item with **bold**, *italic*, and `monospace` inline formatting.

2. Another third-level item containing a math expression: $\sum_{i=1}^n x_i = n(n+1)/2$

2. Back to second level with a bullet sub-list:

- Bullet point A
- Bullet point B with a [hyperlink](#) embedded in text
- Bullet point C

2. Back to first level. This item also contains a footnote.¹

1.3 Mixed List Types in Sequence

- Unordered item 1
- Unordered item 2

1. Ordered item 1
2. Ordered item 2

Term A

Definition of term A with *emphasis*.

Term B

Definition of term B spanning multiple lines to test description list formatting in the converter output.

1.4 Two-Column Layout Section

Column 1: This text appears in the left column of a two-column layout. PDF-to-Markdown converters often struggle with multi-column content because text extraction tools read line-by-line across columns rather than column-by-column. The result is interleaved text from both columns, which produces gibberish unless the converter detects and separates the column flows.

Column 2: The correct behavior is to extract each column's text separately and present them in sequence. Some converters use spatial analysis of bounding boxes to group text into columns before extraction. This paragraph contains enough text to wrap across multiple lines within the column, ensuring the column-detection logic is properly exercised.

2. Typography and Special Characters

2.1 Ligatures and Special Glyphs

The following words contain common ligatures that PDF renderers produce automatically: **field**, **flower**, **affluent**, **official**, **difficult**. These are often mangled during text extraction because the PDF stores them as single glyphs (U+FB01 fi, U+FB02 fl) rather than individual characters.

Additional special characters: em-dash (—), en-dash (–), ellipsis (...), guillemets («French quotes»), and the dreaded \ backslash character that causes escaping nightmares.

2.2 Unicode and Non-ASCII Text

Greek letters inline: α, β, γ, Δ, Ω, λ, μ, σ, π.

Full-width Unicode characters: € (U+20AC), £ (U+00A3), ¥ (U+00A5), © (U+00A9), ® (U+00AE), ™ (U+2122), ° (U+00B0), ± (U+00B1), × (U+00D7), ÷ (U+00F7).

Mathematical operators: $\forall x \in \mathbb{R}, \exists y \ni y > x, \int_0^\infty e^{-x^2} dx = \sqrt{\pi}/2, \nabla \cdot \mathbf{F} = \partial F_x / \partial x + \partial F_y / \partial y.$

2.3 Subscripts, Superscripts, and Chemical Formulas

Temperature ranges: 25°C to 100°C. Concentration: 2.5×10⁶ cells/mL.

The oxidation state of iron in Fe₂O₃ is +3. The nitrate ion is NO₃^{−−}.

Molecular formula with isotope notation: ¹⁴C₆H₁₂O₆.

Chemical equations: 2H₂ + O₂ → 2H₂O CO₂ + Ca(OH)₂ → CaCO₃ + H₂O

²²⁷₉₀Th⁺ SO₄^{2−} + Ba²⁺ → BaSO₄ ↓

2.4 Intra-Word Formatting Changes

This sentence has random font and style changes mid-word to test whether the converter preserves or discards intra-word formatting.

This sentence is entirely red. This one is blue. This is bold AND green.

3. Tabular Data

3.1 Text-Based Table (HTML Table with Merged Cells)

Model	Benchmark Score (%)			Cost per M tokens
	Coding	Math	Reasoning	
Claude Sonnet 5	71.5	89.3	91.1	\$2.00 / \$10.00
Kimi K2.6	61.8	82.1	91.1	\$0.65 / \$3.41
DeepSeek V4 Pro	59.4	85.7	88.8	\$0.44 / \$0.87
Gemini 2.5 Flash	68.2	79.5	84.4	\$0.30 / \$2.50

Table 1: Text-based table with merged header cells. This tests the converter's ability to handle `rowspan` and `colspan` in HTML-derived table structures.

3.2 Rasterized PNG Table

Model	Coding	Math	Reasoning	Cost/M tok
Claude Sonnet 5	71.5%	89.3%	91.1%	2.00/10.00
Kimi K2.6	61.8%	82.1%	91.1%	0.65/3.41
DeepSeek V4 Pro	59.4%	85.7%	88.8%	0.44/0.87
Gemini 2.5 Flash	68.2%	79.5%	84.4%	0.30/2.50

Table 2: Rasterized PNG table. This is an image-based table that requires OCR or vision-model extraction. Converters must detect that this is a table (not a figure) and either extract the structured data or preserve it as an embedded image.

3.3 Rasterized Monitoring Data Table

Station	Temp (°C)	Humidity (%)	Wind (m/s)	Pressure (hPa)	PM2.5
DK001	22.4	65.2	3.1	1013.2	8.4
DK002	23.1	58.7	2.8	1012.8	12.1
DK003	19.8	72.3	4.5	1014.1	6.7
DK004	24.7	55.1	1.9	1011.9	15.3

Table 3: Another rasterized PNG table with numeric alignment. This tests whether the converter can extract right-aligned numeric data from an image-based table.

4. Code and Technical Content

4.1 Python Code

```
def dehyphenate(text: str, next_page: str | None = None) -> str:
    """
    Reassemble hyphenated words split across line and page breaks.

    Edge cases handled:
    - Soft hyphens (U+00AD)
    - Hard hyphens at line endings ← comment with arrow
    - Page-boundary splits
    - Hyphens that are NOT word breaks (e.g., 'state-of-the-art')
    """
    import re

    # Remove soft hyphens unconditionally
    text = text.replace('\xad', '')

    # Line-break hyphens: "implemen-\ntation" → "implementation"
    text = re.sub(r'(\w+)-\n(\w+)', r'\1\2', text)
```

```
# Page-boundary: stitch with next page
if next_page:
    if text.rstrip().endswith('-'):
        text = text.rstrip()[:-1] + next_page.lstrip()

return text
```

Listing 1: Python function with edge-case handling for dehyphenation.

4.2 Shell Command and Output

```
$ df -h /dev/sda1
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       234G  187G   35G   85% /

$ free -m
              total        used        free      shared  buff/cache   available
Mem:          15892         8234         2341          187         5316         7123
Swap:          2048           512         1536
```

Listing 2: Shell commands with output — the `$` prompt and command must be distinguished from the output.

4.3 JSON Data Block

```
{
  "model": "moonshotai/kimi-k2.6",
  "provider": "openrouter",
  "limits": {
    "max_tokens": 65536,
    "requests_per_minute": 60,
    "timeout_seconds": 300
  },
  "features": ["code", "vision", "tools", "structured_output"],
  "pricing": {
    "input_per_mtok": 0.65,
    "output_per_mtok": 3.41,
    "currency": "USD"
  }
}
```

4.4 Verbatim / Pre-formatted Text

```
This is pre-formatted text
with preserved   spacing
and line
breaks.
```

```
def hello():  
    print("Indentation matters!")
```

5. Mathematical and Scientific Content

5.1 Display Equations

The Gaussian integral is one of the most famous results in mathematics:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (1)$$

Equation (1) appears throughout physics and statistics. The multivariate generalization:

$$\int_{\mathbb{R}^n} e^{-(1/2)\mathbf{x}^T \Sigma^{-1} \mathbf{x}} d\mathbf{x} = (2\pi)^{n/2} \sqrt{\det(\Sigma)} \quad (2)$$

5.2 Maxwell's Equations (Aligned)

$$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t \quad (4)$$

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0 \quad (5)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (6)$$

5.3 Matrix Notation

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

6. Figures and Visual Elements

6.1 Figure with Caption

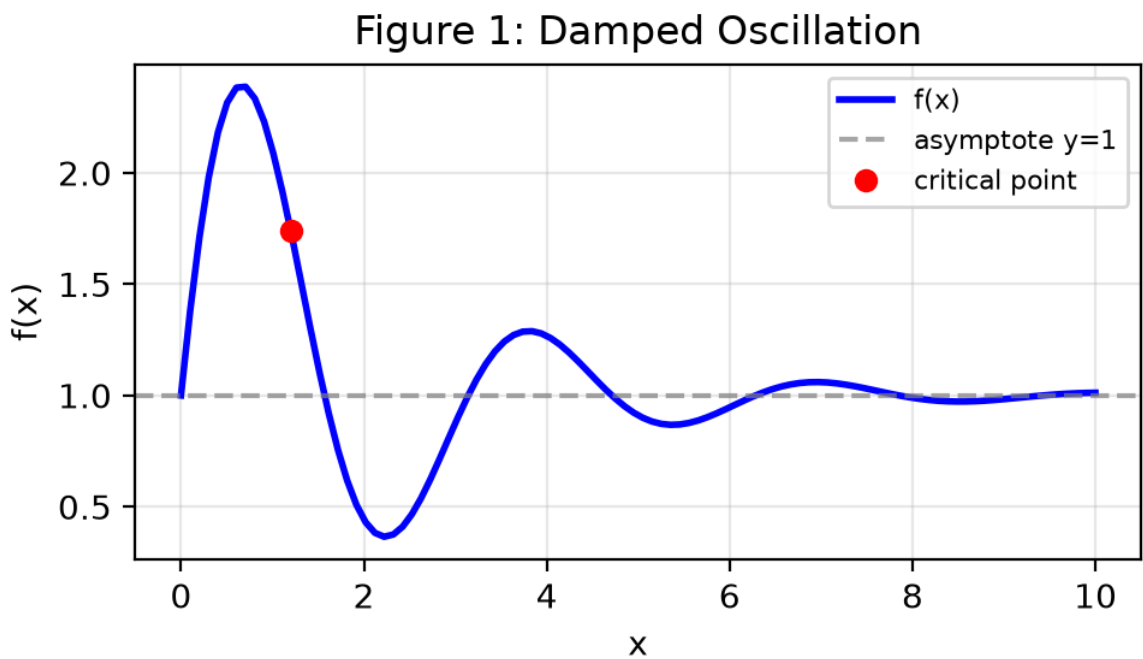


Figure 1: Damped oscillation with annotated features: the blue curve shows $f(x)=2e^{-x/2} \sin(2x)+1$, the dashed line marks the horizontal asymptote, and the red dot indicates a local extremum.

6.2 Wrapped Figure

This paragraph has a figure wrapped to its right side. The text should flow around the figure box rather than being interrupted by it. This is a common layout in technical reports and textbooks where illustrations are placed alongside explanatory text. The converter must recognize that this is a floating element and place it appropriately in the linear Markdown output. Processing large-scale environmental datasets requires careful attention to data quality, temporal consistency, and spatial coverage. The Copernicus Land Monitoring Service produces over 50 datasets at various spatial resolutions ranging from 10 m to 1 km.

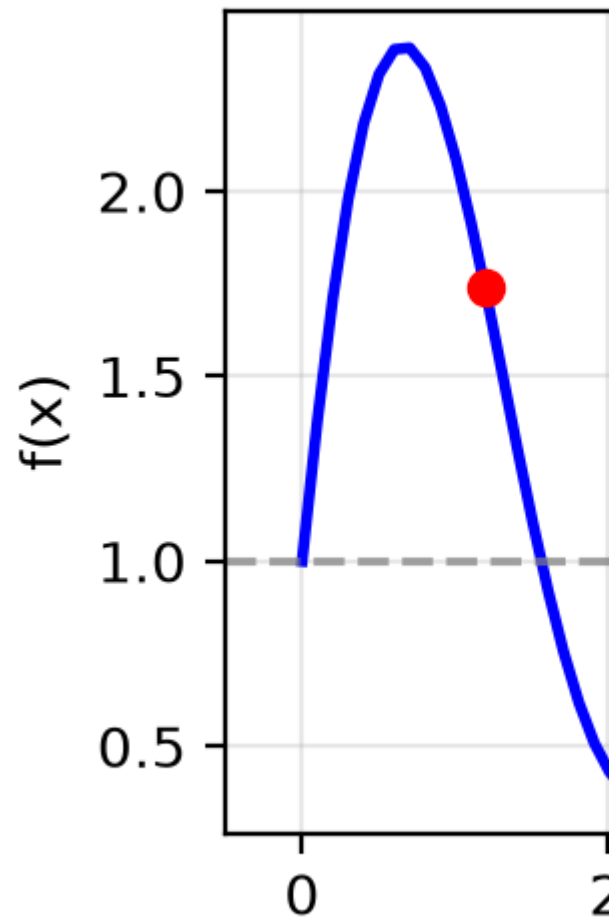


Figure 2: Wrapfigure test

6.3 Subfigures

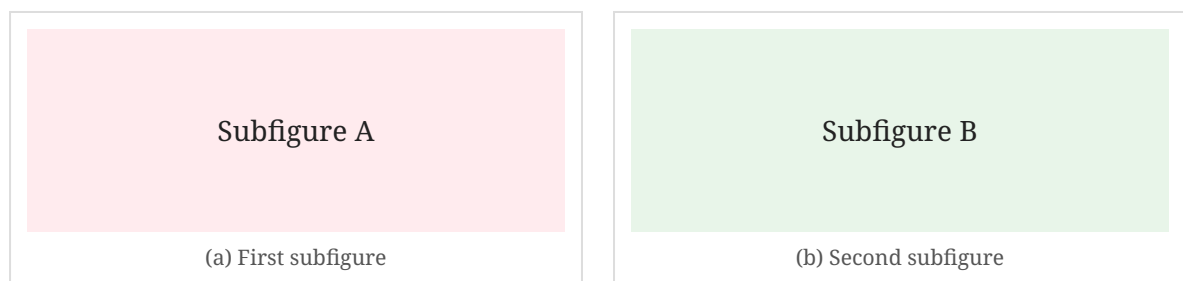


Figure 3: Two subfigures side by side. Subfigure captions must be preserved individually.

6.4 Real-World Example: Copernicus Aerial Photograph



Figure 4: Agricultural area around the village of Gislinge (DK), dominated by small patchy woody features. Source: Copernicus Land Monitoring Service — *Small Woody Features 2015 Product User Manual* (© European Union, 2025). This is a real production document photograph, included to test how the converter handles high-resolution aerial imagery alongside text content.

7. Callout Boxes and Marginal Content

Key Finding: Edge-case handling is the primary determinant of conversion quality, not average-case performance. A model that scores 95% on clean scientific prose but drops footnotes, mangles tables, and loses equation numbering is *worse* than a model scoring 88% that preserves all structural elements.

⚠ **Warning:** Running headers and footers (like "Confidential — Internal Use Only" at the bottom of each page) must be *stripped* from the output. If they appear inline in the converted Markdown, the document structure is corrupted. This is a common failure mode.

8. Footnotes and References

This paragraph has multiple footnotes that test sequential numbering and content preservation.¹ Another sentence with a second footnote.² And a third footnote for good measure.³

8.1 Blockquote with Nested Content

This is a blockquote that may contain multiple paragraphs and nested formatting.

This is a nested blockquote (quote within a quote). It should be rendered with double indentation or double `>` markers in Markdown.

The blockquote also contains a code block:

```
$ curl -s https://api.example.com/v1/status  
{  
  "status": "ok",  
  "uptime": 1234567  
}
```

And an ordered list:

- 1. First quoted item*
- 2. Second quoted item*

8.2 Task Lists and Checkboxes

- ☐ Download raw satellite imagery
- ☒ Pre-process with atmospheric correction
- ☒ Apply cloud mask
- ☐ Compute NDVI
- ☐ Generate final map product

8.3 Definition Lists

NDVI — Normalized Difference Vegetation Index

A spectral index calculated as $(\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$, commonly used to assess vegetation health and density from multispectral satellite imagery.

LAI — Leaf Area Index

Defined as one-half the total green leaf area per unit horizontal ground surface area, typically expressed in m^2/m^2 .

FAPAR — Fraction of Absorbed Photosynthetically Active Radiation

The fraction of incoming solar radiation in the 400–700 nm spectral range that is absorbed by vegetation canopies.

8.4 Horizontal Rules and Section Breaks

Some text before the rule.

Some text after the rule. The converter should render the horizontal rule as `---` or `***` in Markdown.

9. Content at Page Boundaries

This paragraph started on the previous page and continues here after a forced page break. The sentence that was interrupted by the page break must be reassembled correctly. The converter needs to detect that this is a continuation, not a new paragraph. — This is an em-dash-separated clause that might be split across a page boundary, creating ambiguity about whether it's a new sentence or a continuation.

9.1 References

1. Smith, J. and Jones, K. (2025). *Document Structure Analysis: A Survey*. Journal of Document Engineering, 12(3), 234–256. <https://doi.org/10.1000/jde.2025.12.3.234>
2. European Environment Agency (2024). *Copernicus Land Monitoring Service — Product User Manual*. EEA Report No 15/2024. Copenhagen.
3. Černý, P., Novák, M., and Müller, H. (2023). Accented author names and their impact on metadata extraction. In *Proceedings of ICDAR 2023*, pp. 89–97.
4. García-López, A. and Fernández, R. (2026). Tildes, diéresis, and other diacritical marks: a comprehensive test suite for Unicode handling in document conversion. *International Journal of Digital Preservation*, 8(1), 45–67.

¹ First footnote: contains a URL <https://github.com/eea/pdf2md> and some **bold** text.

² Second footnote: multi-paragraph structure. This is the second paragraph of the footnote, which tests whether the converter can handle block-level elements inside footnote content. It also contains a list: (a) Footnote list item 1, (b) Footnote list item 2.

³ Third footnote: `inline code` and *emphasis*.

10 Hyperlink Annotations (Clickable PDF Links)

Tests whether the converter preserves clickable hyperlink annotations. The URLs below are real PDF link annotations, not plain text — the postfix link-recovery step (e29d0b1) reads these from PDF annotation objects, not from visible text.

—
[Click here to visit example.com](#)
 —
contact@example.org
 —
[pdf2md GitHub repository](#)
 —
<https://doi.org/10.5194/essd-14-2785-2022>

11 Typst-Escape Patterns (\DATA, HTML Tags)

Tests `_sanitise_for_typst()` (Phase 2.5c). Bare backslashes before capitals are Typst escapes. Raw `<br>`/`<sub>`/`<sup>` tags are dropped by Typst. All must survive as correct Markdown.

11.1 Bare backslash before capitals:

The `\DATA` field must be preserved. The `\TABLE` reference should also survive. `\TEXT` and `\VALUE` are common. Even `\XYZ` should pass through intact.

Backslash before lowercase stays: `\name`, `\path`.
 Single backslash in LaTeX: `\alpha`, `\beta`, `\gamma`.

11.2 Raw HTML sub/sup/br tags:

The formula`<sub>2</sub>`H`<sub>2</sub>`0 must survive.
 Superscripts: E = mc`<sup>2</sup>`, refs`<sup>[1]</sup>`.

Line breaks: first line`<br>`second line`<br>`third line
`<br>`standalone break

Mixed: H`<sub>2</sub>`S0`<sub>4</sub>` at 25`<sup>°</sup>`C

11.3 Combined:

The `\DATA` field`<sub>raw</sub>` was validated.`<br>`
 The `\TEXT` value was `<sub>nested</sub><sup>[3]</sup>`.

VERIFY: `\\DATA` etc. verbatim in `.qmd`. After sanitiser: backslash+capital → `\\\\DATA`, `<sub>`→`~...~`, `<sup>`→`^...^`, `<br>`→newline.

12 LaTeX Math Delimiters and Ambiguous Notation

Tests (a) whether $...$ display math survives the sanitiser unchanged, and (b) whether ambiguous notation like error bars, thresholds, and set notation is correctly classified as non-math (tightened in commit cd5e9b9).

12.1 Display math (sanitiser must skip $...$):

The Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi}{c^4} T_{\mu\nu}$$

And the fundamental theorem of calculus:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Inline math: $E = mc^2$ is embedded here, and $\alpha = \frac{1}{2}$ appears in prose.

12.2 Ambiguous notation (false-positive guards):

Error bars — must NOT become $...$ math:

Temperature: 23.5 deg C +/- 0.5 deg C (1 sigma)

Pressure: 1013 +/- 15 hPa (95% confidence)

Threshold: values > 2.5 sigma from the mean

Set notation — must NOT become $...$ math:

IDs: {A001, A002, A003} — set of identifiers

Interval: [0, 1], values outside excluded

Coords: (x, y) = (12.5, 34.2) deg N

Statistical expressions in prose:

$p < 0.001$ (statistically significant)

$n = 15$ samples (mean +/- SD)

R-squared = 0.94, adjusted R-squared = 0.91

VERIFY: 12.1 preserves $...$ intact. 12.2 stays plain text, NOT wrapped in $...$. Error bars (+/-), set notation ({...}), and stats ($p < 0.001$) are TEXT — tightened equation detection must reject them.

13 Chrome Variants, Page Numbers, Lint Patterns

This page has a different running header ("Results") than previous pages ("Introduction", "Methods"). The chrome.py module detects per-page header/footer text variants via majority-vote over margin regions. Every supplement page has a "Page N of 3" footer.

13.1 Page-number leakage guard:

The footer on every supplement page reads "Page N of 3" — tests the lint.py chrome-leak guard. Inline references to pages are intentional content (NOT chrome):

```
"see page 12 for the full table"
"continued on page 13"
"as shown on the next page"    # <-- NOT chrome
```

13.2 Nested/relative image paths:

The lint.py module catches nested image markdown — pattern that causes Quarto "File name too long" crashes:

```
BAD (nested — lint must flag):



BAD (invented path — no file exists):
![alt](edgecases-media/image42.jpeg)

```

13.3 Doubled URL schemes:

The linter catches doubled URL schemes:

```
BAD (lint must flag):
https://doi.org/https://doi.org/10.5194/essd-14-2785-2022
https://doi.org/doi.org/10.1234/abc
```

VERIFY: chrome.py detects 3 header variants (Introduction / Methods / Results) + uniform footer. lint.py flags the BAD patterns in 13.2-13.3.